1) P18.7

$(e^{i\theta})^3 = e^{i3\theta} = \cos 3\theta + i\sin 3\theta$

又：$(e^{i\theta})^3 = (\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\sin^2\theta\cos\theta - i\sin^3\theta$

$= (\cos^3\theta - 3\sin^2\theta\cos\theta) + i(3\cos^2\theta\sin\theta - \sin^3\theta)$

∴ $\cos 3\theta = \cos^3\theta - 3\sin^2\theta\cos\theta$

2) P18.9

$P(z_0)=0$ $\cancel{z_0\bar{z}_0 = |z_0|^2}$ 设 $z_0 = x_0 + iy_0$

$\cancel{P(\bar{z}_0) =}$ $P(z_0)=0$, $z^n + \bar{z}_0^n = A$ $(A\in R)$

$P(z_0)+P(\bar{z}_0) = B$ $(B\in R)$

又：$z^n - \bar{z}^n = iC$ $(C\in R)$

∴ $P(z_0)-P(\bar{z}_0) = iD$ $(D\in R)$

∴ $P(\bar{z}_0) = B = -iD$ $(B.D\in R)$

∴ $B=D=0 \Rightarrow P(\bar{z}_0)=0$

3) P18.10.

1) $z^4 = -1 = e^{\pi+2k\pi}$ $k\in Z$

∴ $z = e^{\frac{\pi}{4}+\frac{1}{2}k\pi} = e^{\frac{\pi}{4}} / e^{\frac{3}{4}\pi} / e^{\frac{5}{4}\pi} / e^{\frac{7}{4}\pi}$

2) $(z+i)^5 = 1 = e^{2k\pi}$ $k\in Z$

∴ $z+i = e^{\frac{2}{5}k\pi}$

$z = e^{\frac{2}{5}k\pi} - i = 1-i / e^{\frac{2}{5}\pi}-i / e^{\frac{4}{5}\pi}-i / e^{\frac{6}{5}\pi}-i / e^{\frac{8}{5}\pi}-i$

④ P18, 11

$w^n = 1$，要证 $w^0 + w + w^2 + \cdots + w^{n-1} = 0$

~~即证：$w^0 + w^1 + \cdots + w^{n-1} + w^n = 1$~~

设 $w^0 + w^1 + \cdots + w^{n-1} = A$

$wA = w^1 + w^2 + \cdots + w^{n-1} + w^n = 1 + w^1 + \cdots w^{n-1} = A$

$\therefore (w-1)A = 0$

$\therefore A = 0$，得证

⑤ P47, 5

~~$z = x + iy$~~

~~$LHS = \lim\limits_{\substack{x\to 0\\ y\to 0}} \frac{e^x(\cos y + i\sin y) - 1}{x + iy}$~~

(1) $\lim\limits_{z\to 0} \frac{e^z - 1}{z} = \lim\limits_{z\to 0} e^z = 1$

(2) $\lim\limits_{z\to 0} \frac{\sin|z|}{z} = \begin{cases} \lim\limits_{z\to 0^+} \frac{\sin z}{z}, & z > 0 \\ \lim\limits_{z\to 0^-} \frac{-\sin z}{z}, & z < 0 \end{cases}$

当 $z > 0$. $LHS = \lim\limits_{z\to 0^+} \cos z = 1$

同理 $z < 0$. $LHS = 1$

$\therefore Ans = 1$

⑥ P48, 8.

(2) $f(z) = x^2 + iy^2$.

① ~~可导：f(z)~~ $\frac{\partial u}{\partial x} = 2x \quad \frac{\partial u}{\partial y} = 0$

$\frac{\partial v}{\partial x} = 0$. $\frac{\partial v}{\partial y} = 2y$

$\therefore$ 在 $x = y$ 上，$f(z)$ 可导，不可解析

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(3) $\cancel{f(z)=}$ $\frac{\partial u}{\partial x}=\cos x\,\mathrm{ch}y$ $\frac{\partial u}{\partial y}=\sin x\,\mathrm{sh}y$

$\frac{\partial v}{\partial x}=-\sin x\,\mathrm{sh}y$ $\frac{\partial v}{\partial y}=\cos x\,\mathrm{ch}y$

$\therefore f(z)$在C上可导可解析

7. P49.18

(4) 设$z=x+iy$

$$e^{\frac{1}{z}}=e^{\frac{x-iy}{x^2+y^2}}=e^{\frac{x}{x^2+y^2}-i\frac{y}{x^2+y^2}}=\underbrace{e^{\frac{x}{x^2+y^2}}\cos\left(-\frac{y}{x^2+y^2}\right)}_{Re(e^{\frac{1}{z}})}+i\underbrace{e^{\frac{x}{x^2+y^2}}\sin\left(-\frac{y}{x^2+y^2}\right)}_{Im(e^{\frac{1}{z}})}$$

8. P49.20

(3) $\ln z=\frac{\pi}{2}i$

有: $\mathrm{Ln}z=\ln z+i2k\pi$, $k'=-k$

$\therefore e^{\ln z}=e^{\mathrm{Ln}z+i2k'\pi}=e^{i\frac{\pi}{2}} \Rightarrow z\cdot e^{i\cdot 2k'\pi}=e^{i\frac{\pi}{2}}$

$\therefore z=e^{i(\frac{\pi}{2}-2k'\pi)}=e^{i(\frac{\pi}{2}+2k\pi)}=i$